CASE STUDY · OCTOBER 2026

Expectation vs. Reality

Three Jenbacher J320 gensets after 303 days of operation

96.3%
of hours running
(94.5% expected)

84.7%
average output while running
(95% expected)

−8.5%
modelled year-1 EBITDA
vs. recalculated study

33.6%
modelled pre-tax IRR, local currency
(40.1% expected)

Igor Shibanov, CEO K4-TECH

1 · THE PROMISE

The study promised a 2.4-year payback at 95% load

Physics

- 3 × J320, 3,189 kW installed
- Availability 0.95; output while running 95%
- 8,125 h per unit per year (92.8%)
- 24.6 GWh/yr at 0.271 Nm³/kWh

Prices (excl. VAT)

- Gas USD 0.088/Nm³
- Electricity value USD 102/MWh
- Heat USD 6.2/Gcal, 58% used

Money

- CAPEX USD 3.5M (USD 1,110/kW)
- EBITDA USD 1.49M/yr
- Payback 2.38 yr = CAPEX ÷ EBITDA
- No IRR calculated

We recalculated the study on the investor-agreed basis: commissioning 1 Oct 2025, 5-year horizon, 10% indexation, 10% discount rate, 20% profit tax, 10-year depreciation, prices net of 18% VAT, major overhaul at 80,000 h.

> The study's EBITDA mixes 10-year average revenue with costs from another period and costs gas at full load for every running hour.

Hours, kWh and stop notes are verifiable. Gas basis and heat use are not.

1 Jun 2025
construction start

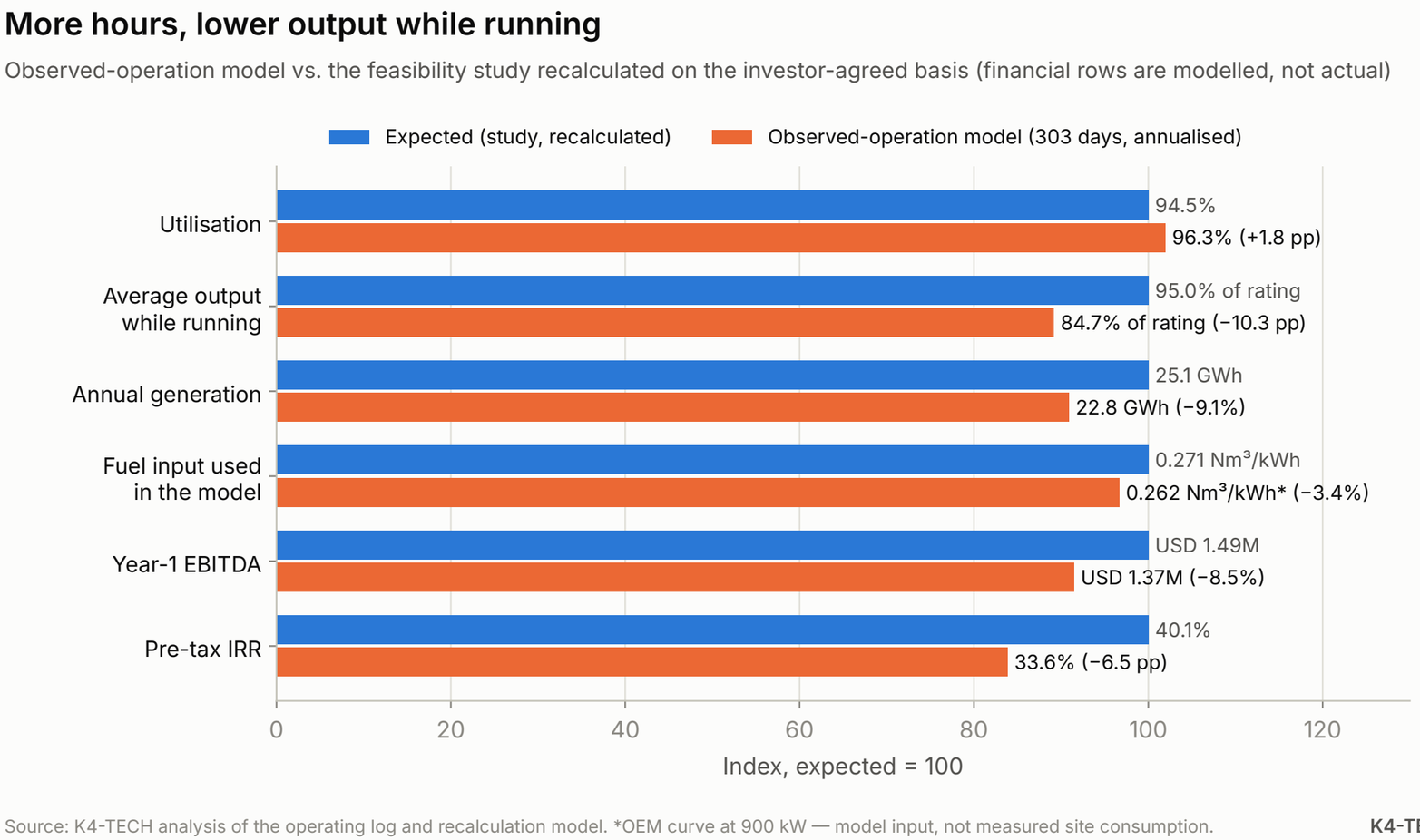
1 Oct 2025
commissioning

2 Nov 2025 – 31 Aug 2026
daily log: 303 dates × 3 units = 21,816 unit-hours

30 Sep 2030
end of 5-year horizon

Data	Source	Status
Running hours, kWh	Daily operating log	Verified
Stop causes	Log notes	Partial — 44% blank
Gas consumption	Manufacturer curve (site meter unverified)	Benchmark
Oil condition	18 lab samples	Verified (aggregate)
Prices 2025–26	Investor	Reported
Heat use, first month	—	Missing

More hours, lower output while running



> Fuel input is the manufacturer curve at 900 kW — a model input, not measured site consumption. Financial rows are modelled (investor-agreed basis), not actual site results.

4 · UTILISATION VS. OUTPUT WHILE RUNNING

The engines ran more than planned — but delivered less while running. Two different numbers.

96.3%

Utilisation
21,008.5 of 21,816 unit-hours
expected 94.5% (study model) · 92.8% (original)

900 kW

Average output while running
84.7% of the 1,063 kW rating
expected 1,010 kW (95%)

A model or contract that uses one "availability" figure for both will be wrong in both directions. We avoid "load factor", which engineers often apply to the whole period including stops.

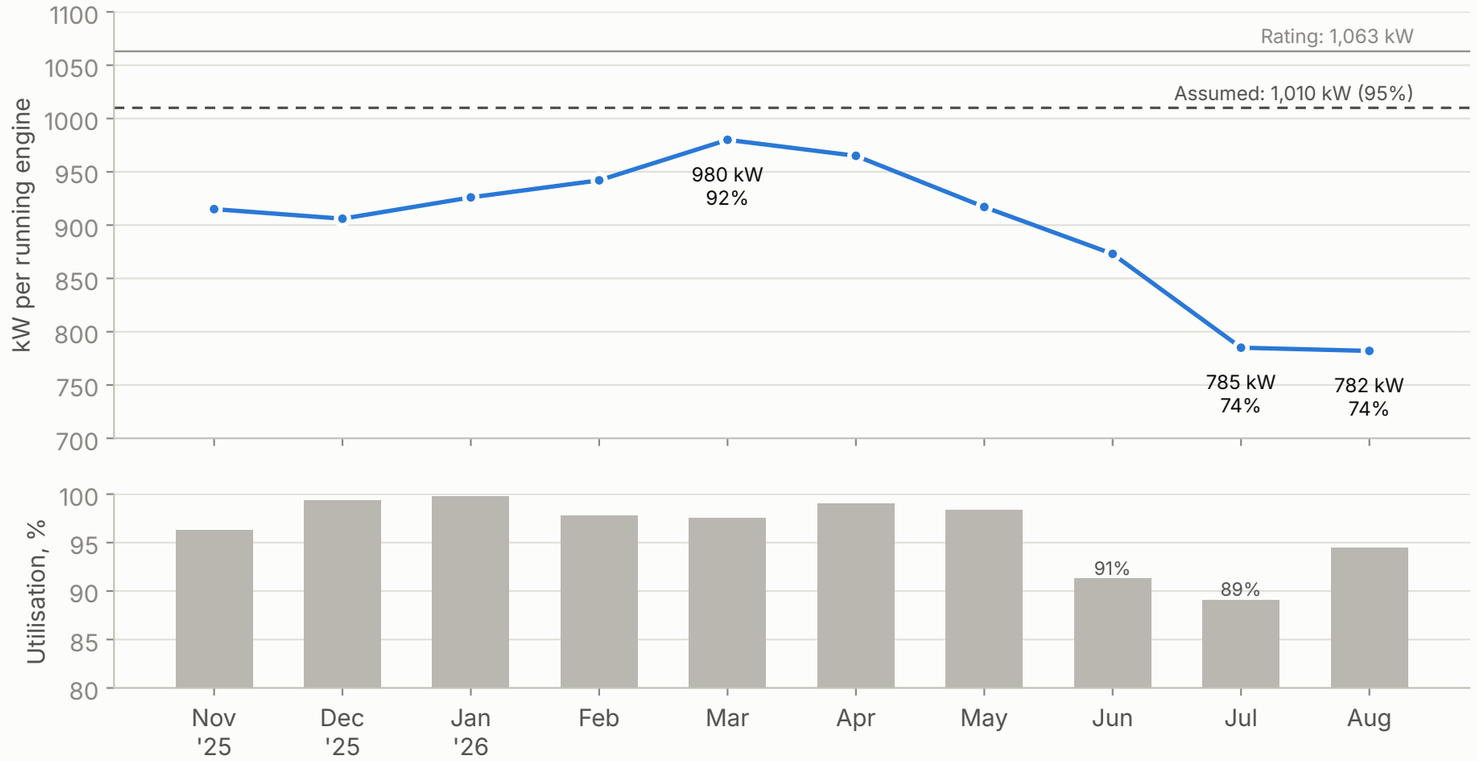
Correction: an earlier K4-TECH case page showed 96.5% / 759 lost hours, combining two date bases. Consistent bases give 96.3% / 807.5 h; the case page is updated.

5 · MONTHLY OUTPUT

The plant kept running, but at part load in summer

The plant kept running, but at part load

Output per running engine fell from 92% of rating in March to 74% in July–August; running hours barely changed



Source: K4-TECH analysis of the plant operating log (2 Nov 2025–31 Aug 2026) and the K4-TECH recalculation model.

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Operator-reported causes: outdoor temperature and lower consumer load; the split is not measured. kWh metering point not confirmed.

6 · GAS

The manufacturer curve is the defensible modelling basis — not measured site consumption

0.271

Nm³/kWh
Feasibility study
(full-load gas × every running hour)

0.262

Nm³/kWh
Manufacturer curve at ~900 kW
(reference LHV 9.5 kWh/Nm³) — used in the model

0.240

m³/kWh
Site meter — **unverified**
would imply ~44% electrical efficiency

The meter trend is still useful: gas per kWh rose about a fifth from winter to July as load fell, as part-load operation predicts.

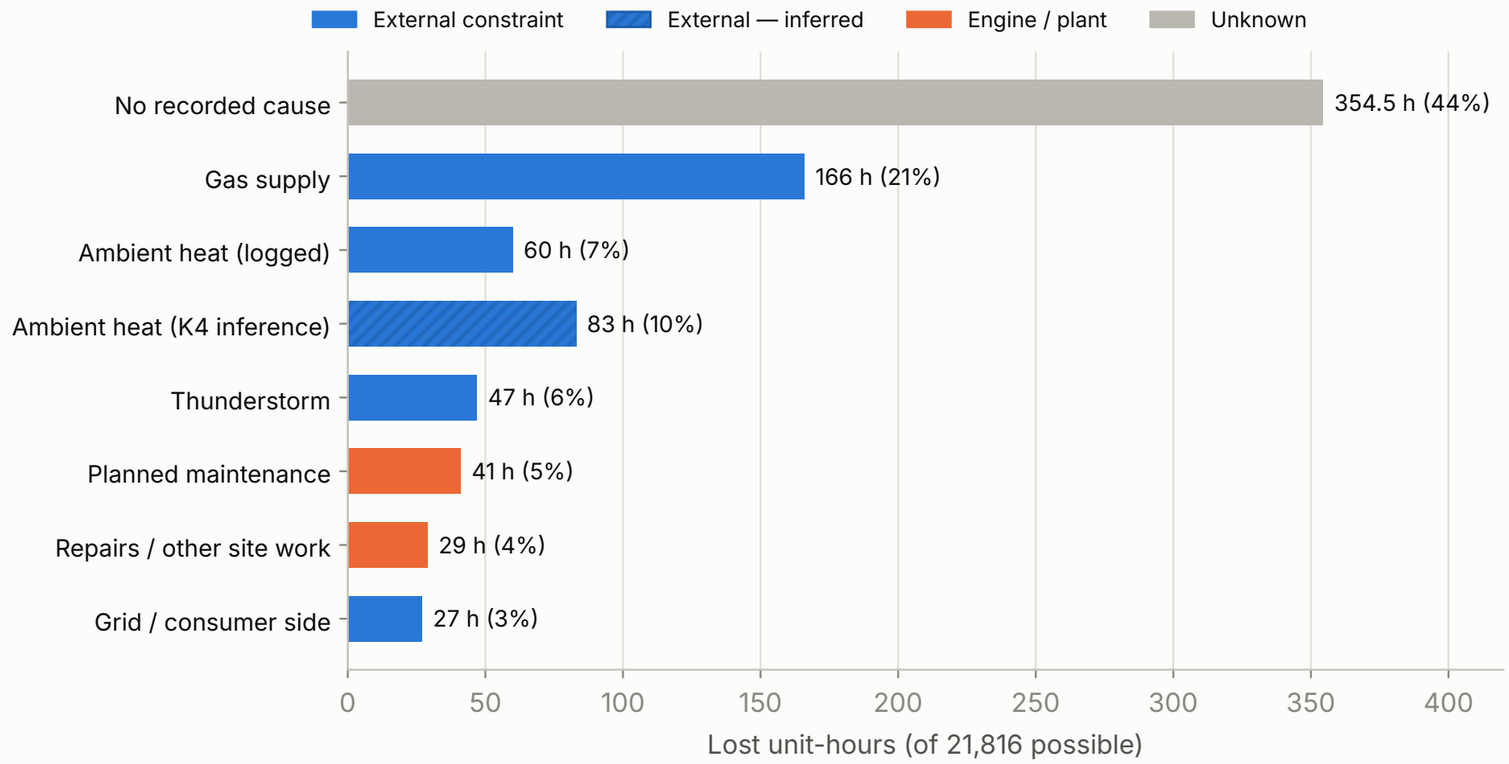
> "The operating log implies approximately 0.240 m³/kWh, but the metering basis and reference conditions have not been verified."

7 · DOWNTIME

Recorded external constraints cost under 2% of hours; 44% of lost hours have no cause

What stopped the plant: 807.5 lost unit-hours

External constraints: 300–383 h (1.4–1.8% of possible hours). 44% of lost hours have no recorded cause



Source: K4-TECH analysis of the plant operating log (2 Nov 2025–31 Aug 2026) and the K4-TECH recalculation model.

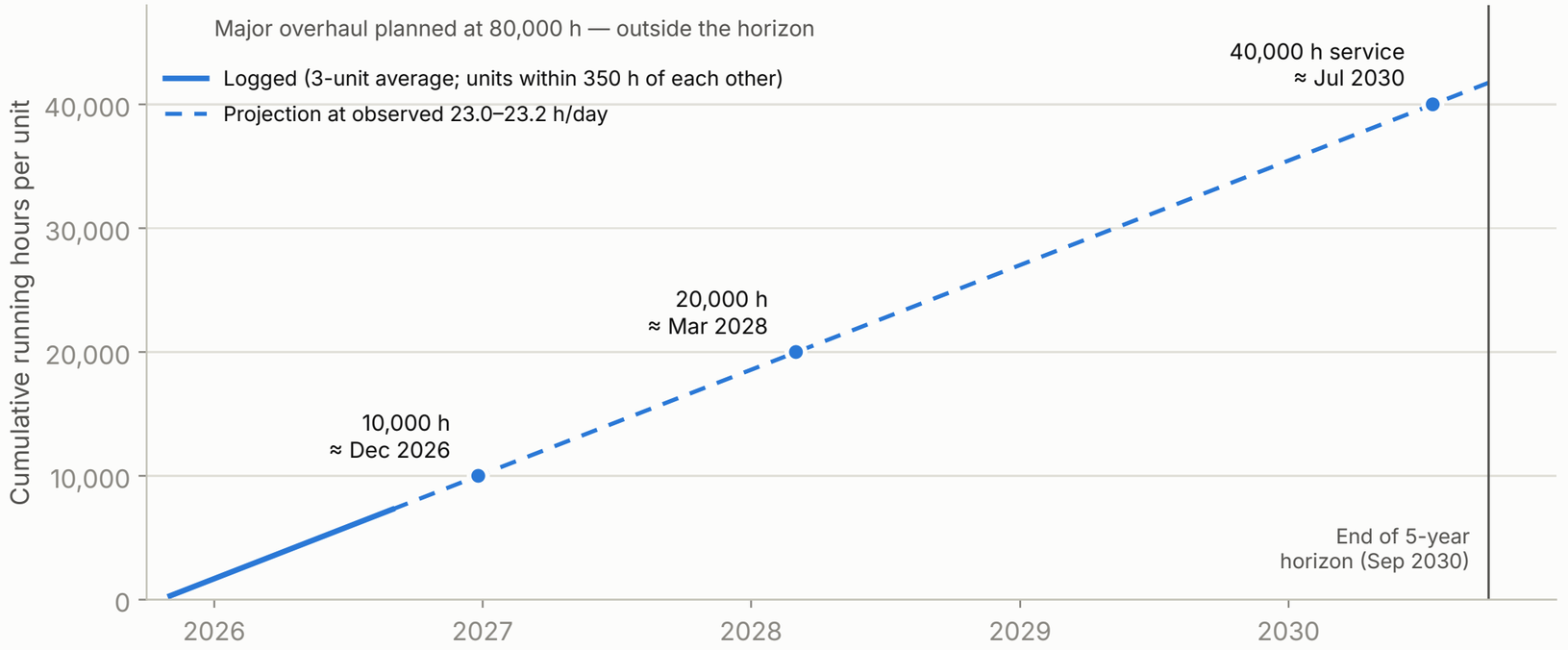
K4-TECH

83 h of heat are K4-TECH inference (same daytime pattern, no note). Modelled value of external downtime: about USD 23–29k a year at investor-reported prices.

At ~8,400 h a year, the 40,000-hour service lands in year 5

Maintenance follows running hours, not the calendar

At ~8,400 h per unit a year, the 40,000-hour service falls around July 2030 — inside a 5-year horizon



Source: plant operating log (cumulative hours, 2 Nov 2025–31 Aug 2026); K4-TECH linear projection.

K4-TECH

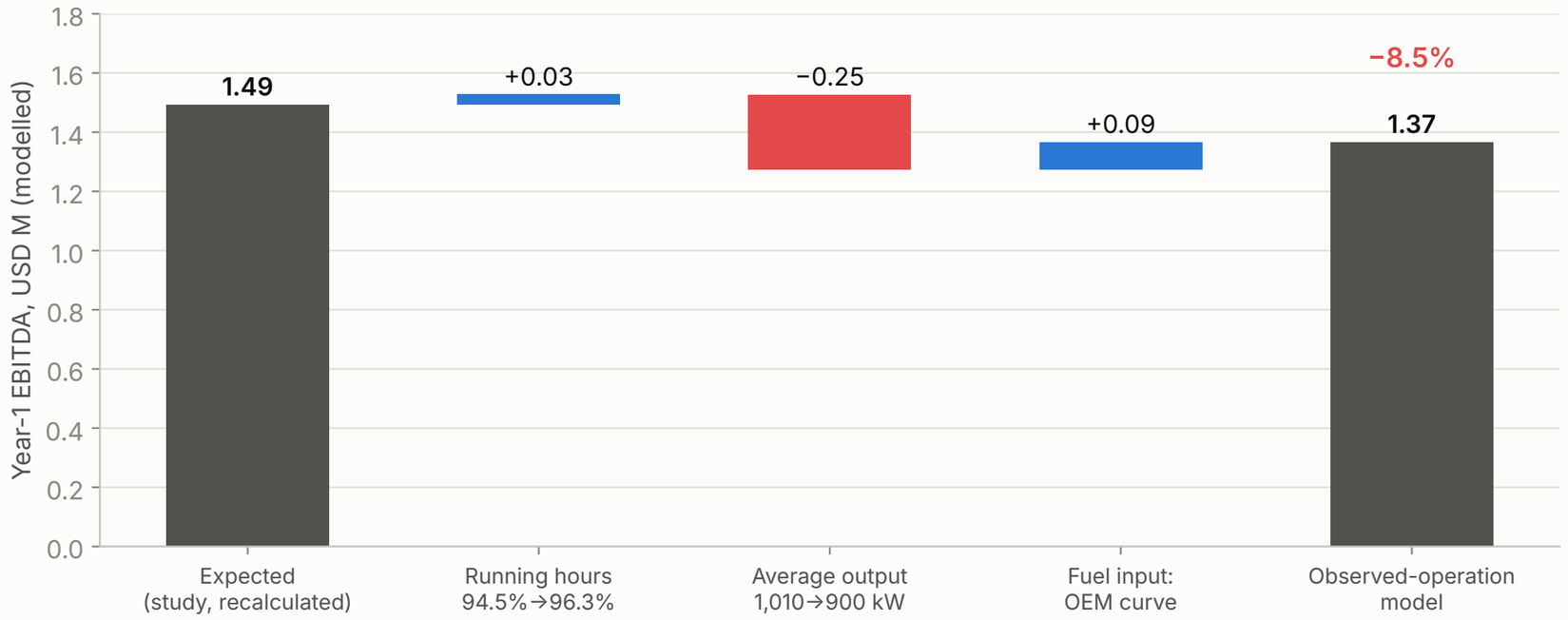
Major overhaul planned at 80,000 h (outside the horizon). Scope at 40,000 h moves IRR between 30.2% and 34.4%. No maintenance entry in the log after 6,666 h.

9 · EBITDA BRIDGE

On one price basis, lower output while running drives the −8.5%

From expectation to the observed-operation model: year-1 EBITDA

Same prices and finance basis on both ends; only hours, output and fuel input change. Modelled — not actual site EBITDA



Source: K4-TECH recalculation model, single-input steps; investor-agreed price basis. USD at 80.5307 per USD (official central-bank rate, 28 Oct 2026). **K4-TECH**

> Modelled — not actual site EBITDA. At the original study prices the same steps give USD 1.71M → 1.56M (−8.7%); 1.71 vs 1.49 is price basis only.

10 · FIVE-YEAR ECONOMICS

Observed-operation model: about 6½ points less IRR, 0.4 years longer payback

	Expected study, recalculated	Observed-operation model
Year-1 EBITDA	USD 1.49M	USD 1.37M
5-year EBITDA	USD 7.96M	USD 7.20M
IRR pre-tax*	40.1%	33.6%
IRR post-tax* — investor-agreed tax basis	29.1%	23.7%
IRR post-tax* — statutory-rate scenario	24.1%	19.1%
NPV @ 10%, pre-tax	USD 2.6M	USD 2.0M
Payback from commissioning	2.3 yr	2.7 yr

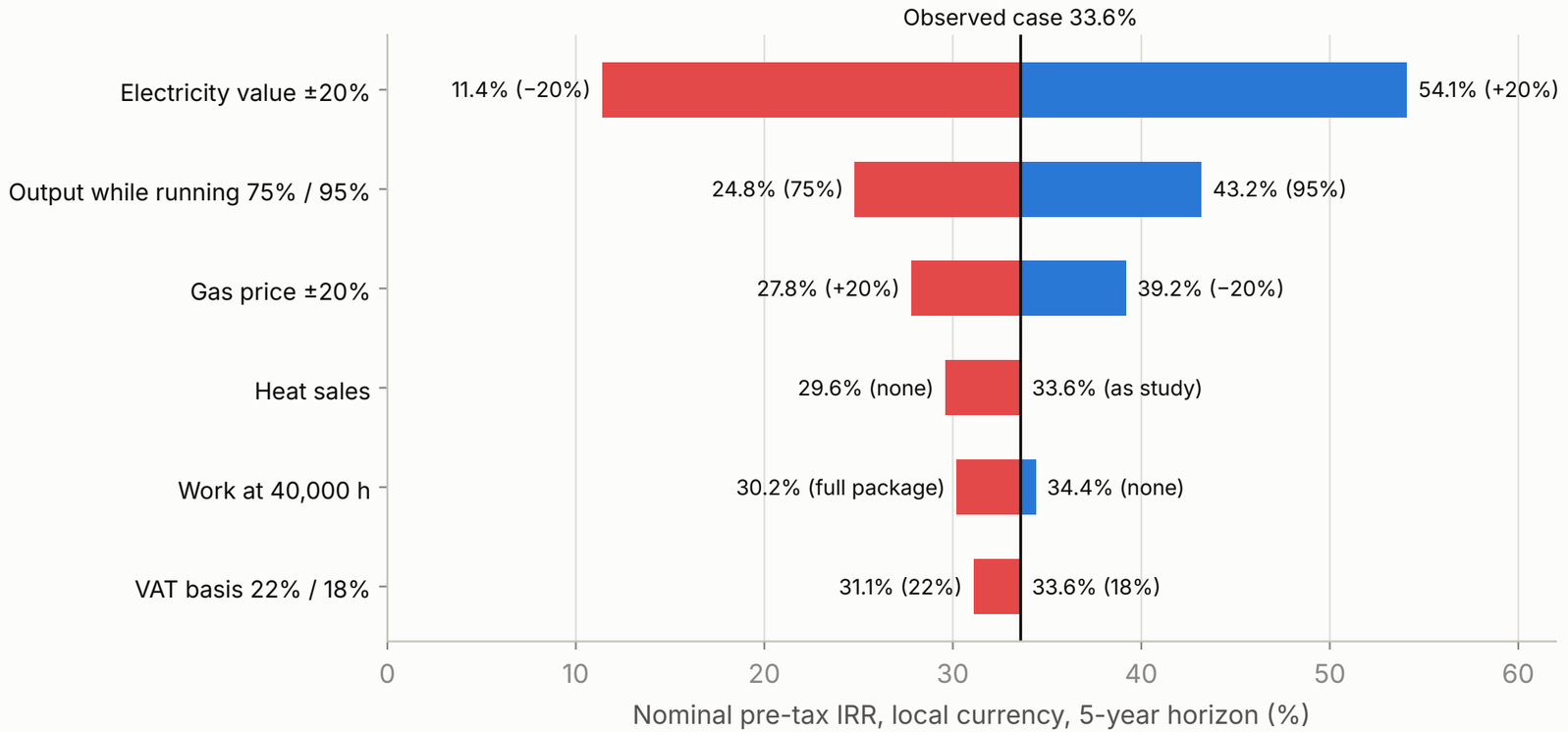
*Nominal local-currency IRR with 10% indexation — not comparable with a USD hurdle rate. Investor-agreed basis: 18% VAT, 20% profit tax; statutory-rate scenario uses the higher rates now in force. Modelled, not actual. Zero residual value after 5 years.

11 · SENSITIVITY

Electricity value and output while running move the model most

What moves the modelled return

In this case, electricity value and output while running move the model more than recorded engine-related downtime



Source: K4-TECH recalculation model, observed-operation case; one input changed at a time. Local-currency IRR is not comparable with a USD hurdle rate. **K4-TECH**

> Reliability was not tested as a separate variable; 354.5 lost hours have no recorded cause. One input changed at a time.

12 · DECISION CHECKLIST

Eight questions to ask before you sign

1. Show me a daily log from a comparable plant — hours, kWh, gas and a cause for every stop.
2. What is the summer output at my site's temperatures, and what cooling is assumed?
3. In which cubic metres is gas metered, at what heating value, and where is the meter?
4. Is maintenance placed by running hours? When does the 40,000-hour service fall, and who pays?
5. Which kWh price is used — tariff, avoided cost or mining equivalent — and with or without VAT?
6. Who carries gas-supply, grid and consumer-side stops in the contract?
7. Pre-tax or post-tax? Which tax rates actually in force, discount rate, depreciation and residual value?
8. Is the IRR in local currency or in USD?

K4-TECH reviews feasibility studies and financial models on the payer's side — investor, lender, owner. **k4-technology.com** · linkedin.com/in/igor-shibanov-k4